

are the inverse system to $(a, b, c, d, f, g, h, l, m, n)$, then let

$$\begin{aligned}\alpha_1 &= \mathfrak{A}\alpha + \mathfrak{H}\beta + \mathfrak{G}\gamma + \mathfrak{L}\delta, \\ \beta_1 &= \mathfrak{H}\alpha + \mathfrak{B}\beta + \mathfrak{F}\gamma + \mathfrak{M}\delta, \\ \gamma_1 &= \mathfrak{G}\alpha + \mathfrak{F}\beta + \mathfrak{C}\gamma + \mathfrak{N}\delta, \\ \delta_1 &= \mathfrak{L}\alpha + \mathfrak{M}\beta + \mathfrak{N}\gamma + \mathfrak{D}\delta,\end{aligned}$$

and in like manner,

$$\begin{aligned}\xi_1 &= \mathfrak{A}\xi + \mathfrak{H}\eta + \mathfrak{G}\zeta + \mathfrak{L}\omega, \\ \eta_1 &= \mathfrak{H}\xi + \mathfrak{B}\eta + \mathfrak{F}\zeta + \mathfrak{M}\omega, \\ \zeta_1 &= \mathfrak{G}\xi + \mathfrak{F}\eta + \mathfrak{C}\zeta + \mathfrak{N}\omega, \\ \omega_1 &= \mathfrak{L}\xi + \mathfrak{M}\eta + \mathfrak{N}\zeta + \mathfrak{D}\omega,\end{aligned}$$

then the coordinates of the pole of the fixed plane are as

$$\alpha_1 : \beta_1 : \gamma_1 : \delta_1,$$

and the coordinates of the pole of the variable plane are as

$$\xi_1 : \eta_1 : \zeta_1 : \omega_1,$$

whence the equations of the polar are

$$\begin{vmatrix} x, & y, & z, & w \\ \alpha_1, & \beta_1, & \gamma_1, & \delta_1 \\ \xi_1, & \eta_1, & \zeta_1, & \omega_1 \end{vmatrix} = 0,$$

a system of equations which may be thus represented

$$\begin{aligned}\xi_1 &= K\lambda x + \mu\alpha_1, \\ \eta_1 &= K\lambda y + \mu\beta_1, \\ \zeta_1 &= K\lambda z + \mu\gamma_1, \\ \omega_1 &= K\lambda w + \mu\delta_1,\end{aligned}$$

where K is the discriminant of the system

$$(a, b, c, d, e, f, g, h, l, m, n).$$

Write

$$\begin{aligned}x &= ax + hy + gz + lw, \\ y &= hx + by + fz + mw, \\ z &= gx + fy + cz + nw, \\ w &= lx + my + nz + dw,\end{aligned}$$

the last preceding system of equations may be written

$$\begin{aligned}\xi &= \lambda x + \mu\alpha, \\ \eta &= \lambda y + \mu\beta, \\ \zeta &= \lambda z + \mu\gamma, \\ \omega &= \lambda w + \mu\delta,\end{aligned}$$

equations in which λ, μ are indeterminate, and where x, y, z, w may be considered as current coordinates, and this system represents the polar above referred to. Combining the equations in question with the equation

$$\xi x + \eta y + \zeta z + \omega w = 0,$$

of the variable plane, we have

$$\lambda(\alpha x + \beta y + \gamma z + \delta w) + \mu(\alpha x + \beta y + \gamma z + \delta w) = 0,$$

i.e.

$$\lambda(a, \dots)(x, y, z, w)^2 + \mu(\alpha x + \beta y + \gamma z + \delta w) = 0,$$

or what is the same thing

$$\lambda : \mu = \alpha x + \beta y + \gamma z + \delta w : (a, \dots)(x, y, z, w)^2,$$

and substituting these values in the expressions for ξ , η , ζ , ω we have ξ , η , ζ , ω in terms of the coordinates x , y , z , w of the pole above referred to, i.e., if for shortness,

$$U = (a, b, c, d, f, g, h, l, m, n)(x, y, z, w)^2,$$

$$P = \alpha x + \beta y + \gamma z + \delta w,$$

then

$$\xi = \frac{1}{2}P d_x U - \alpha U,$$

$$\eta = \frac{1}{2}P d_y U - \beta U,$$

$$\zeta = \frac{1}{2}P d_z U - \gamma U,$$

$$\omega = \frac{1}{2}P d_w U - \delta U,$$

and combining with these equations the equation

$$(\mathfrak{a}, \dots)(\xi, \eta, \zeta, \omega)^2 = 0,$$

we have

$$(\mathfrak{a}, \dots)\left(\frac{1}{2}P d_x U - \alpha U, \frac{1}{2}P d_y U - \beta U, \frac{1}{2}P d_z U - \gamma U, \frac{1}{2}P d_w U - \delta U\right)^2 = 0,$$

for the required locus of the pole of the line of intersection of the variable plane and the fixed plane, with the conic of intersection of the given surface of the second order and the variable plane. The locus in question is a surface of the fourth order; and it may be remarked that this surface touches the given surface of the second order along the conic of intersection with the fixed plane.

7th April, 1857.

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NOTE ON THE ‘CIRCULAR RELATION’ OF PROF. MÖBIUS.

[*From the Quarterly Mathematical Journal*, vol. II. (1858), p. 162.]

THEOREM.

Given the points

$$A, B, C; P,$$

and the points

$$A', B', C';$$

describe the circles α , β , γ , ω as follows: viz.

$$\alpha \text{ through } (B, C, P),$$

$$\beta \quad \text{„} \quad (C, A, P),$$

$$\gamma \quad \text{„} \quad (A, B, P),$$

$$\omega \quad \text{„} \quad (A, B, C),$$

and the circles $\alpha', \beta', \gamma', \omega'$ as follows: viz.,

$$\begin{aligned} &\alpha' \text{ through } (B', C') \text{ cutting } \omega' \text{ at the angle at which } \alpha \text{ cuts } \omega, \\ &\beta' \quad \text{,, } (C', A') \text{ ,, } \omega' \quad \text{,, } \quad \text{,, } \beta \text{,, } \omega, \\ &\gamma' \quad \text{,, } (A', B') \text{ ,, } \omega' \quad \text{,, } \quad \text{,, } \gamma \text{,, } \omega, \text{ and} \\ &\omega' \quad \text{,, } (A', B', C') \end{aligned}$$

then will α', β', γ' meet in a point P' , i.e. we shall have the points A', B', C' ;

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such that the circles $\alpha', \beta', \gamma', \omega'$ pass

$$\begin{aligned} &\alpha' \text{ through } (B', C', P'), \\ &\beta' \quad \text{,, } (C', A', P'), \\ &\gamma' \quad \text{,, } (A', B', P'), \\ &\omega' \quad \text{,, } (A', B', C'). \end{aligned}$$

We may construct in this manner two figures, such that to three points of the first figure there correspond in the second figure three points which may be taken at pleasure, but these once selected to every other point of the first figure there will correspond in the second figure a perfectly determinate point. And the two figures will be such that whenever in the first figure four or more points lie in a circle, then in the second figure the corresponding points will also lie in a circle. The relation in question is due to Prof. Möbius, who has termed it the *Kreis-verwandschaft* (circular relation) of two plane figures. See his paper *Crelle*, t. LII. [1856], pp. 218–228, extracted from the *Berichte* of the Royal Saxon Society of Sciences of the 5th Feb. 1853, [and *Werke*, t. II. pp. 207–217].

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ON THE DETERMINATION OF THE VALUE OF A CERTAIN DETERMINANT.

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 163–166.]

CONSIDERING the determinant

$$\begin{vmatrix} \theta, & 1, & \cdot & \cdots \\ n, & \theta, & 2 & \\ \cdot & n-1, & \theta, & 3 \\ \cdot & \cdot & n-2, & \theta, & 4 \\ \cdot & \cdot & \cdot & \cdots \end{vmatrix}$$

let the successive diagonal minors be $U_0, U_1, U_2, \dots, U_x, \dots$, it is easy to find

$$\begin{aligned} U_0 &= 1, \\ U_1 &= \theta, \\ U_2 &= \theta^2 - n, \\ U_3 &= \theta(\theta^2 - 4) - 3(n-2)\theta, \\ U_4 &= (\theta^2 - 1)(\theta^2 - 9) - 6(n-3)(\theta^2 - 1) + 3(n-3)(n-1), \end{aligned}$$

which in fact suggests the law, viz.

$$\begin{aligned}
 U_x &= (\theta + x - 1)(\theta + x - 3)(\theta + x - 5) \dots (\theta - x + 5)(\theta - x + 3)(\theta - x + 1) \\
 &\quad - \frac{x(x-1)}{2}(n-x+1)(\theta+x-3)(\theta+x-5) \dots (\theta-x+3) \\
 &\quad + \frac{x(x-1)(x-2)(x-3)}{2 \cdot 4}(n-x+1)(n-x+3)(\theta+x-5) \dots (\theta-x+5) \\
 &\quad - \&c. \\
 &\quad \vdots \\
 &\quad + (-)^s \frac{x(x-1) \dots (x-2s+1)}{2 \cdot 4 \dots 2s}(n-x+1)(n-x+3) \dots (n-x+2s-1) \\
 &\quad \times (\theta+x-2s-1)(\theta+x-2s-3) \dots (\theta-x+2s+1)
 \end{aligned}$$

to $s = \frac{1}{2}x$ or $\frac{1}{2}(x-1)$, as x is even or odd.

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And of course if x denote the number of lines or columns of the determinant, then U_x is the value of the determinant. This theorem, or a particular case of it, is due to Prof. Sylvester: I have not been able to find an easier demonstration than the following one, which, it must be admitted, is somewhat complicated. I observe that U_x satisfies the equation

$$U_x - \theta U_{x-1} + (x-1)(n-x+2)U_{x-2} = 0.$$

Hence writing $x-1$ and $x-2$ for x , we have the system

$$\begin{aligned}
 U_x - \theta U_{x-1} + (x-1)(n-x+2)U_{x-2} &= 0, \\
 U_{x-1} - \theta U_{x-2} + (x-2)(n-x+3)U_{x-3} &= 0, \\
 U_{x-2} - \theta U_{x-3} + (x-3)(n-x+4)U_{x-4} &= 0,
 \end{aligned}$$

or, eliminating U_{x-1} and U_{x-3} ,

$$U_x + \{(x-1)(n-x+2) + (x-2)(n-x+3) - \theta^2\}U_{x-2} + (x-2)(x-3)(n-x+3)(n-x+4)U_{x-4} = 0.$$

Suppose, for shortness,

$$(\theta + x - 1)(\theta + x - 3)(\theta + x - 5) \dots (\theta - x + 5)(\theta - x + 3)(\theta - x + 1) = H_x,$$

and assume

$$U_x = A_{x,1}H_x - A_{x,1}H_{x-2} \dots + (-)^s A_{x,s}H_{x-2s} \dots,$$

where $A_{x,s}$ is independent of θ , then

$$U_x \text{ contains the term } (-)^s A_{x,s}H_{x-2s},$$

$$U_{x-2} \text{ contains the term } (-)^s A_{x-2,s}H_{x-2s-2},$$

which is to be multiplied by

$$(x-1)(n-x+2) + (x-2)(n-x+3) - \theta^2.$$

This multiplier may be written under the form

$$\begin{aligned}
 &(x-1)(n-x+2) + (x-2)(n-x+3) - (x-2s-1)^2 - \{\theta^2 - (x-2s-1)^2\} \\
 &= M_{x,s} - \{\theta^2 - (x-2s-1)^2\},
 \end{aligned}$$

if, for shortness,

$$M_{x,s} = (x-1)(n-x+2) + (x-2)(n-x+3) - (x-2s-1)^2.$$

Now

$$M_{x,s} - \{\theta^2 - (x-2s-1)^2\}$$

multiplied into

$$(-)^s A_{x-2,s}H_{x-2s-2}$$

gives rise to the terms

$$(-)^s M_{x,s} A_{x-2,s} H_{x-2s-2} - (-)^s A_{x-2,s} H_{x-2s},$$

(since $\{\theta^2 - (x - 2s - 1)^2\} H_{x-2s-2} = H_{x-2s}$), or, what is the same thing,

$$\begin{aligned} &= (-)^s M_{x,s} A_{x-2,s} H_{x-2s-2} - (-)^s A_{x-2,s} H_{x-2s} \\ &= -(-)^{s+1} \{M_{x,s} A_{x-2,s} H_{x-2s-2} + A_{x-2,s} H_{x-2s}\}, \end{aligned}$$

and moreover

$$U_{x-4} \text{ contains the term } (-)^s A_{x-4,s-1} H_{x-2s-2},$$

or, what is the same thing, $(-)^s A_{x-4,s-1} H_{x-2s-2}$.

Hence we must have

$$A_{x,s} - (A_{x-2,s} + M_{x,s-1} A_{x-2,s-1}) + (x-2)(x-3)(n-x+3)(n-x+4) A_{x-4,s-1} = 0,$$

where

$$M_{x,s-1} = (x-1)(n-x+2) + (x-2)(n-x+3) - (x-2s+1)^2.$$

This may be satisfied by assuming

$$A_{x,s} = -B_{x,s} (n-x+3)(n-x+5) \dots (n-x+2s+1);$$

for then

$$\begin{aligned} A_{x-2,s} &= B_{x-2,s} (n-x+3) \dots (n-x+2s-1) \dots (n-x+2s+1), \\ A_{x-2,s-1} &= B_{x-2,s-1} (n-x+3) \dots (n-x+2s-1), \\ (n-x+3)(n-x+4) A_{x-4,s-1} &= -B_{x-4,s-1} (n-x+4)(n-x+3) \dots (n-x+2s-1), \end{aligned}$$

and consequently

$$\begin{aligned} &B_{x,s} (n-x+1) \\ &- B_{x-2,s} (n-x+2s+1) \\ &- B_{x-2,s-1} M_{x,s-1} \\ &+ B_{x-4,s-1} (x-2)(x-3)(n-x+4) = 0; \end{aligned}$$

and if this equation be satisfied independently of n , we must have

$$\begin{aligned} B_{x,s} - B_{x-2,s} - (2x-3) B_{x-2,s-1} + (x-2)(x-3) B_{x-4,s-1} &= 0, \\ B_{x,s} - (2x+1) B_{x-2,s} - \{5x-8 - (x-2s+1)^2\} 2s + 4(x-2)(x-3) B_{x-4,s-1} &= 0, \end{aligned}$$

and these are both satisfied by

$$B_{x,s} = \frac{x \cdot x-1 \dots x-2s+1}{2 \cdot 1 \cdot 2 \cdot 3 \dots s},$$

in fact, substituting this value and omitting the factor

$$\frac{(x-2)(x-3) \dots (x-2s+1)}{2 \cdot 1 \cdot 2 \cdot 3 \dots s},$$

the first equation becomes

$$x(x-1) - (x-2s)(x-2s-1) - (2s-3)2s + 4s(s-1) = 0,$$

and the second equation becomes

$$x(x-1) - (2s+1)(x-2s)(x-2s-1) - \{5x-8 - (x-2s+1)^2\} 2s + 16s(s-1) = 0,$$

which are each of them an identical equation, the first being

$$\begin{aligned} & x^2 - x \\ & - x^2 + (4s + 1)x - 2s(2s + 1) \\ & - 4sx + 6s \\ & + 4s(s - 1) = 0, \end{aligned}$$

and the second being

$$\begin{aligned} & x^2 - x \\ & - (2s + 1)\{x^2 - (4s + 1)x + 2s(2s + 1)\} \\ & + 2s\{x^2 - (4s + 3)x + (2s - 1)^2 + 8\} \\ & + 16s(s - 1) = 0, \end{aligned}$$

as may be easily verified.

Hence writing for $B_{x,s}$ its value and recapitulating, the equation

$$U_x + \{(x - 1)(n - x + 2) + (x - 2)(n - x + 3) - \theta^2\} U_{x-2} + (x - 2)(x - 3)(n - x + 3)(n - x + 4) U_{x-4} = 0$$

is satisfied by

$$U_x = A_{x,1}H_x - A_{x,1}H_{x-2} \dots + (-)^s A_{x,s}H_{x-2s} \dots \text{ to } s = \frac{1}{2}x \text{ or } \frac{1}{2}(x - 1), \text{ as } x \text{ is even or odd,}$$

where

$$\begin{aligned} H_x &= (\theta + x - 1)(\theta + x - 3)(\theta + x - 5) \dots (\theta - x + 5)(\theta - x + 3)(\theta - x + 1), \\ A_{x,s} &= \frac{x(x - 1) \dots (x - 2s + 1)}{2 \cdot 1 \cdot 2 \cdot 3 \dots s} (n - x + 1)(n - x + 3) \dots (n - x + 2s - 1), \end{aligned}$$

and since for $x = 0, 1, 2, 3$ the values of the expression U_x coincide with those of the first four diagonal minors, the expression gives in general the value of the diagonal minor, or when x denotes the number of lines or columns of the determinant, then the value of the determinant.

2, *Stone Buildings*, 1st April, 1857.

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ON THE SUMS OF CERTAIN SERIES ARISING FROM THE EQUATION $x = u + tfx$.

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 167–171.]

LAGRANGE has given the following formula for the sum of the inverse n th powers of the roots of the equation $x = u + tfx$,

$$\Sigma(x^{-n}) = u^{-n} + (-nu^{-n-1}fu)\frac{t}{1} + (-nu^{-n-1}f^2u)\frac{t^2}{1 \cdot 2} + \&c. \quad (1)$$

where n is a positive integer and the series on the second side of the equation is to be continued as long as the exponent of u remains negative (*Théorie des Équations Numériques*, p. 225). Applying this to the equation $x = 1 + tx^s$, we have

$$\Sigma(x^{-n}) = 1 - \frac{n}{1}t \cdot 1^{-n+s-1} + \frac{n(n-2s+1)}{1 \cdot 2}t^2 \cdot 1^{-n+2s-1} \dots + (-)^q \frac{n(n-qs+q-1) \dots (n-qs+q+1)}{1 \cdot 2 \dots q} t^q \cdot 1^{-n+qs-q} - \&c. \quad (2)$$

to be continued while the exponent of 1 remains negative.

Let $n = \mu + p$, μ being not greater than $s - 1$, the series may always be continued up to $q = \mu$, and no further. In fact writing the above value for n and putting $q = p + \theta$, the general term is

$$(-)^{p+\theta} \frac{t^{p+\theta}}{1 \cdot 2 \dots (\mu + \theta)} (\mu + p)(\mu + p - \theta s + \theta - 1) \dots (\mu + p - \theta s + \theta + 1) \cdot 1^{-\mu+p-\theta s+\theta}.$$

Now if $p + \mu - \theta(s - 1)$ is negative or zero, the term is to be rejected on account of the index of 1 not being negative, and if this quantity be positive, then since

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$p - \theta s + 1$ is necessarily negative for any value of θ greater than zero, the factorial $(p - \theta s + \mu + \theta - 1) \dots (p - \theta s + 1)$ begins with a positive and ends with a negative factor, and since the successive factors diminish by unity, one of them is necessarily equal to zero, or the term vanishes; hence the series is always to be continued up to $q = \mu$.

Hence

$$\Sigma(x^{-\mu-p}) = 1 - \frac{\mu+p}{1}t + \frac{(\mu+p)(\mu-2s+\rho+1)}{1 \cdot 2}t^2 \dots + (-)^\mu \frac{(\mu+p)(\mu-qs+p+q+1) \dots (\mu-qs+p+1)}{1 \cdot 2 \dots q}t^\mu - \&c. \quad (3)$$

continued to $q = \mu$.

By taking the terms in a reverse order, it is easy to derive

$$(-)^\mu t^{-\mu} \Sigma(x^{-\mu-p}) = (\mu+p) \left\{ \frac{(\mu+p-1) \dots (\mu+1)}{2 \cdot 3 \dots p} + (-)^q \frac{(\mu+p)(\mu+p-1) \dots (\mu+1-q)}{2 \cdot 3 \dots (p+q) \cdot s} t^{-q} - \&c. \right\} \quad (4)$$

continued to $q = \mu$.

Suppose in particular $s = 2$, and $t = -\frac{a+1}{a^2}$, so that the equation in x becomes $\frac{1}{x^2} = \frac{a+1}{x} - \frac{a}{x^2}$, whence $x = a$ or substituting in (2), we find

$$\Sigma\left(\frac{1}{x^n}\right) - \frac{a}{a^n} = 1 + \frac{n}{1} \frac{a+1}{a^2} + \frac{n(n-3)}{2} \frac{(a+1)^2}{a^4} + \&c. \quad (5)$$

continued to the term involving $\left(\frac{a+1}{a^2}\right)^{\frac{1}{2}n}$ or $\left(\frac{a+1}{a^2}\right)^{\frac{1}{2}(n-1)}$.

Put $a = -\frac{b}{a}$ and therefore

$$a+1 = \frac{b-a+1}{a} = \frac{b}{a} \cdot \frac{a+1}{a}, \quad \frac{a+1}{a^2} = \frac{a+b}{a^2} \cdot \frac{a+1}{a^2} = \frac{ab}{(a-b)^2},$$

we obtain

$$\frac{a^n + b^n}{(a+b)^n} = 1 - \frac{n}{1} \frac{ab}{(a+b)^2} + \frac{n(n-3)}{1 \cdot 2} \frac{a^2 b^2}{(a+b)^4} - \&c., \quad (6)$$

or

$$\frac{(a+b)^n - a^n - b^n}{nab(a+b)} = (a+b)^{n-2} - \frac{n-3}{1 \cdot 2} (a+b)^{n-4} ab + \frac{(n-4)(n-5)}{2 \cdot 3} (a+b)^{n-6} a^2 b^2 - \&c. \quad (7)$$

to be continued as long as the exponent of $(a+b)$ on the second side is negative.

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This formula, which is easily deducible from that for the expansion of $\cos n\theta$ in powers of $\cos \theta$, is employed by M. Stern, *Crelle*, t. XX. [1840], in proving the following theorem: If

$$S = 1 - \frac{n-3}{2} + \frac{(n-4)(n-5)}{2 \cdot 3} - \&c. \quad (8)$$

continued to the first term that vanishes, then according as n is of the form $6k+3$, $6k \pm 1$, $6k$ or $6k \pm 2$,

$$S = \frac{2}{n}, \quad S = 0, \quad S = -\frac{1}{n}, \quad S = \frac{1}{n}, \quad (9)$$

which is in fact immediately deduced from it by writing $b = \omega a$, ω being one of the impossible cube roots of unity. Substituting the above values of x in the equation (4),

$$(1+a)^{p+1} - (1+a)^{-p} = (2p+1)a \left\{ 1 + \frac{(p+1)p}{2 \cdot 3} \frac{a^2}{a+1} + \frac{(p+2)(p+1)p(p-1)}{2 \cdot 3 \cdot 4 \cdot 5} \frac{a^4}{(a+1)^2} + \dots \right\}, \quad (10)$$

$$(1+a)^p + (1+a)^{-p} = 2p \left\{ \frac{1}{p} + \frac{p}{2} \frac{a^2}{a+1} + \frac{(p+1)p(p-1)}{2 \cdot 3 \cdot 4} \frac{a^4}{(a+1)^2} + \dots \right\}, \quad (11)$$

whence

$$(1+a)^{p+1} + (1+a)^{-p} = (2p+1)a \left\{ 1 + \frac{(p+1)p}{2 \cdot 3} \frac{a^2}{a+1} + \dots \right\} + 2p \left\{ \frac{1}{p} + \frac{p}{2} \frac{a^2}{a+1} + \dots \right\} = U \text{ suppose,} \quad (12)$$

i.e.

$$\Delta(-)^p(1+a)^p = (-)^{p+1}U \text{ or } (1+a)^p = (-)^p \Sigma(-)^{p+1}U,$$

where Δ and Σ refer to the variable p . The summation is readily effected by means of the formulae

$$\Sigma(-)^{p+1}(2p+1)(p+r+1) \dots (p-s) = (-)^p(p+r+1) \dots (p-s-1),$$

$$\Sigma(-)^{p+1}(p+s) \dots (p-s) 2p = (-)^p(p+s) \dots (p-s-1),$$

and we thence find

$$(1+a)^p = \left\{ 1 + \frac{p(p-1)}{1 \cdot 2} \frac{a^2}{1+a} + \frac{(p+1)p(p-1)(p-2)}{1 \cdot 2 \cdot 3 \cdot 4} \frac{a^4}{(1+a)^2} + \dots \right\} + \frac{a}{1+a} \left\{ p + \frac{(p+1)p(p-1)}{1 \cdot 2 \cdot 3} \frac{a^2}{1+a} + \dots \right\}. \quad (13)$$

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a formula of Euler's (*Pet. Trans.* 1811) demonstrated likewise by M. Catalan (*Liouville*, t. IX. [1844], pp. 161–174) by induction. It may be expressed also in the slightly different form

$$(1+a)^p = \left\{ 1 + \frac{(p+1)p}{1 \cdot 2} \frac{a^2}{1+a} + \frac{(p+2)(p+1)p(p-1)}{1 \cdot 2 \cdot 3 \cdot 4} \frac{a^4}{(1+a)^2} + \dots \right\} + \frac{a}{1+a} \left\{ p + \frac{(p+1)p(p-1)}{1 \cdot 2 \cdot 3} \frac{a^2}{1+a} + \dots \right\}. \quad (14)$$

The two series (13), (14) are each of them supposed to contain $p+1$ terms, p being an integer; but since the terms after these all of them vanish, the series may be continued indefinitely. Suppose the two sides expanded in powers of p , the coefficients will be separately equal, and thus the identity of the two sides will be independent of the particular values of p , or the equations (13), (14), and similarly, (10), (11), (12) are true for any values of p whatever. It is to be observed that the series for negative values of p do not differ essentially from those for the corresponding positive values; as may be seen immediately by writing $-p$ for p , and $\frac{-a}{1+a}$ for a .

Suppose next $s = 3$, or that the equation in x is $x = 1 + tx^3$; to rationalise the roots of this, assume $t = \frac{4(\beta^2 - 1)}{(\beta^2 + 3)^3}$, then values of x are

$$x = \frac{\beta^2 + 3}{2(\beta + 1)}, \quad x = -\frac{\beta^2 + 3}{2(\beta - 1)}, \quad x = \frac{\beta^2 + 3}{\beta^2 - 1},$$

and hence

$$\frac{2^n \{ (\beta + 1)^n + (-)^n (\beta - 1)^n \} + (\beta^2 - 1)^n}{(\beta^2 + 3)^n} = 1 - \frac{n}{1} t + \frac{n(n-5)}{1 \cdot 2} t^2 - \dots + (-)^q \frac{n(n-2r-1) \dots (n-3r+1)}{1 \cdot 2 \dots r} t^r + \&c. \quad (15)$$

where $t = \frac{4(\beta^2 - 1)}{(\beta^2 + 3)^3}$, and the series is to be continued up to the term involving $t^{\frac{1}{2}n}$, $t^{\frac{1}{2}(n-1)}$ or $t^{\frac{1}{2}(n-2)}$.

Again, from the formula (4) we deduce the three following forms,

$$\frac{(-)^p 2^{p+1} \{ (\beta + 1)^{p+1} + (-)^p (\beta - 1)^{p+1} \} + (\beta^2 - 1)^{p+1}}{2^p (\beta^2 - 1)^p} = (3\mu + 1) \left\{ \frac{1}{\mu} + \frac{(\mu+1)p}{2 \cdot 3} t^{-1} + \frac{(\mu+2)(\mu+1)p(\mu+p-1)}{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} t^{-2} + \dots + (-)^q \frac{(\mu+3q-1) \dots (\mu-q+1)}{2 \cdot 3 \dots 3q} t^{-q} \right\}, \quad (16)$$

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$$\frac{(-)^p 2^{p+1} \{ (\beta + 1)^{p+1} - (-)^p (\beta - 1)^{p+1} \} + (\beta^2 - 1)^{p+1}}{2^p (\beta^2 - 1)^p (\beta^2 + 3)} = (3\mu + 1) \left\{ 1 - \frac{(\mu+2)(\mu+1)\mu}{2 \cdot 3 \cdot 4} t^{-1} + \dots + (-)^q \frac{(\mu+2q) \dots (\mu-q+1)}{2 \cdot 3 \dots 3q+1} t^{-q} + \dots \right\}, \quad (17)$$

$$\frac{(-)^p 2^{p+1} \{ (\beta + 1)^{p+1} + (-)^p (\beta - 1)^{p+1} \} + (\beta^2 - 1)^{p+1}}{2^p (\beta^2 - 1)^p (\beta^2 + 3)} = (3\mu + 2) \left\{ \frac{\mu+1}{2} + \frac{(\mu+3)(\mu+2)(\mu+1)\mu}{2 \cdot 3 \cdot 4 \cdot 5} t^{-1} + \dots + (-)^q \frac{(\mu+2q+1) \dots (\mu-q+1)}{2 \cdot 3 \dots (3q+2)} t^{-q} + \dots \right\}, \quad (18)$$

all of them continued up to $q = \mu$.

2, Stone Buildings, 1st April, 1857.

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ON THE SIMULTANEOUS TRANSFORMATION OF TWO HOMO-
GENEOUS FUNCTIONS OF THE SECOND ORDER.

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 192–195.]

IN a former paper with this title, *Cambridge and Dublin Math. Journal*, t. IV. [1849], pp. 47–50 [74], I gave (founded on the methods of Jacobi and Prof. Boole) a simple solution of the problem, but the solution may I think be presented in an improved form as follows, where as before I consider for greater convenience the case of three variables only.

Suppose that by the linear transformation (†)

$$(x, y, z) = \begin{pmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{pmatrix} (x_1, y_1, z_1),$$

we have identically

$$(a, b, c, f, g, h \mid x, y, z)^2 = (a_1, b_1, c_1, f_1, g_1, h_1 \mid x_1, y_1, z_1)^2, \\ (A, B, C, F, G, H \mid x, y, z)^2 = (A_1, B_1, C_1, F_1, G_1, H_1 \mid x_1, y_1, z_1)^2,$$

and write also

$$(\xi, \eta, \zeta) = \begin{pmatrix} \alpha, & \alpha', & \alpha'' \\ \beta, & \beta', & \beta'' \\ \gamma, & \gamma', & \gamma'' \end{pmatrix} (\xi_1, \eta_1, \zeta_1)$$

and so in all like cases.

Comparing these with the relations between (x, y, z) and (x_1, y_1, z_1) , we see that

$$(\xi, \eta, \zeta \mid x, y, z) = (\xi_1, \eta_1, \zeta_1 \mid x_1, y_1, z_1),$$

and multiplying the first of the relations between two quadrics by an indeterminate quantity λ , and adding it to the second, we have

$$(\lambda a + A, \dots \mid x, y, z)^2 = (\lambda a_1 + A_1, \dots \mid x_1, y_1, z_1)^2.$$

We have thus a linear function and a quadric transformed into functions of the same form by means of the linear substitutions, and any invariant of the system will remain unaltered to a factor *près*, such factor being a power of the determinant of substitution. The invariants are, 1° the discriminant of the quadric; 2° the reciprocant, considered not as a contravariant of the quadric, but as an invariant of the system. And if we write

$$K = \text{Disc.} (\lambda a + A, \dots \mid x, y, z)^2, \\ (\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H} \mid \xi, \eta, \zeta)^2 = \text{Recip.} (\lambda a + A, \dots \mid x, y, z)^2,$$

then K_1 , &c. being the analogous expressions for the transformed functions, and the determinant of substitution being represented by Π , we have

$$K_1 = \Pi^2 K, \\ (\mathfrak{A}_1, \dots \mid \xi_1, \eta_1, \zeta_1)^2 = \Pi^2 (\mathfrak{A}, \dots \mid \xi, \eta, \zeta)^2,$$

and substituting for ξ_1, η_1, ζ_1 their values in terms of ξ, η, ζ , the last equation breaks up into six equations, and we have

$$(\mathfrak{A}_1, \dots \mid \alpha, \alpha', \alpha'')^2 = \Pi^2 \mathfrak{A},$$

⁰† I represent in this manner the system of equations $x = \alpha x_1 + \beta y_1 + \gamma z_1$, &c.

$$(\mathfrak{A}_1, \dots | \alpha\beta, \alpha'\beta', \alpha''\beta'')(\gamma, \gamma', \gamma'') = \Pi^2 \mathfrak{H},$$

which is the system obtained in a somewhat different manner in my former paper. Putting $f_1 = g_1 = h_1 = F_1 = G_1 = H_1 = 0$, and writing also (which is no additional loss of generality) $a_1 = b_1 = c_1 = 1$, the formulas become

$$(a, b, c, f, g, h | x, y, z)^2 = (1, 1, 1 | x_1^2, y_1^2, z_1^2),$$

$$(A, B, C, F, G, H | x, y, z)^2 = (A_1, B_1, C_1 | x_1^2, y_1^2, z_1^2),$$

viz. there are two given quadrics which are to be by the same linear substitution transformed, one of them into the form $x_1^2 + y_1^2 + z_1^2$ and the other into the form $A_1 x_1^2 + B_1 y_1^2 + C_1 z_1^2$, where A_1, B_1, C_1 have to be determined. The solution is contained in the following system of formulæ, viz.

$$(A_1 + \lambda)(B_1 + \lambda)(C_1 + \lambda) = \Pi^2 \text{Disc.}(\lambda a + A, \dots).$$

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which gives A_1, B_1, C_1 as the roots of a cubic equation, and gives also

$$1 = \Pi^2 \text{Disc.}(a, \dots) = \Pi^2 \kappa, \text{ or } \Pi^2 = \frac{1}{\kappa} \text{ suppose,}$$

and we have then, writing for shortness, $(* | X, Y, Z)^2$ for

$$((B_1 + \lambda)(C_1 + \lambda), (C_1 + \lambda)(A_1 + \lambda), (A_1 + \lambda)(B_1 + \lambda) | X, Y, Z),$$

$$(* | \alpha^2, \alpha'^2, \alpha''^2) = \frac{1}{\kappa} \mathfrak{A},$$

$$(* | \beta^2, \beta'^2, \beta''^2) = \frac{1}{\kappa} \mathfrak{B},$$

$$(* | \gamma^2, \gamma'^2, \gamma''^2) = \frac{1}{\kappa} \mathfrak{C},$$

$$(* | \beta\gamma, \beta'\gamma', \beta''\gamma'') = \frac{1}{\kappa} \mathfrak{F},$$

$$(* | \gamma\alpha, \gamma'\alpha', \gamma''\alpha'') = \frac{1}{\kappa} \mathfrak{G},$$

$$(* | \alpha\beta, \alpha'\beta', \alpha''\beta'') = \frac{1}{\kappa} \mathfrak{H},$$

where $(\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H})$ are the coefficients of the reciprocant of $(\lambda a + A, \dots | x, y, z)^2$. Writing $\lambda = -A_1, -B_1$, or $-C_1$, the quadric functions on the left-hand side become mere monomials, and we have the actual values of the squares and products $\alpha^2, \beta\gamma$, &c. of the coefficients of the linear substitutions: thus $\alpha^2, \beta^2, \gamma^2, \beta\gamma, \gamma\alpha, \alpha\beta$ are respectively equal to $\mathfrak{A}_*, \mathfrak{B}_*, \mathfrak{C}_*, \mathfrak{F}_*, \mathfrak{G}_*, \mathfrak{H}_*$, each into the common factor

$$\frac{1}{\kappa}(B_1 - A_1)(C_1 - A_1),$$

the suffix denoting that we are to write in the expressions for $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}$ the value $-A_1$ for λ ; and similarly for the sets $(\alpha', \beta', \gamma')$ and $(\alpha'', \beta'', \gamma'')$.

2, *Stone Buildings, 27th March, 1857.*

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NOTE ON A FORMULA IN FINITE DIFFERENCES.

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 198–201.]

IN Jacobi's Memoir "De usu legitimo formulae summatoriae Maclauriniana," *Crelle*, t. XII. [1834], pp. 263–273 (1834), expressions are given for the sums of the odd powers of the natural numbers $1, 2, 3 \dots x$ in terms of the quantity

$$u = x(x + 1),$$

viz. putting for shortness

$$Sx^r = 1^r + 2^r + \dots + x^r,$$

the expressions in question are

$$\begin{aligned} Sx^3 &= \frac{1}{2}u^2, \\ Sx^5 &= \frac{1}{3}u^2(u - \frac{1}{2}), \\ Sx^7 &= \frac{1}{6}u^2(u^2 - \frac{4}{3}u + \frac{2}{3}), \\ Sx^9 &= \frac{1}{5}u^2(u^3 - 3u^2 + 3u - \frac{3}{5}), \\ Sx^{11} &= \frac{1}{2}u^2(u^4 - 4u^3 + \frac{27}{2}u^2 - 10u + 5), \\ Sx^{13} &= \frac{1}{4}u^2(u^5 - \frac{5}{3}u^4 + \frac{25}{3}u^3 - 11u^2 + \frac{691}{60}u - \frac{691}{105}), \end{aligned}$$

&c., which, especially as regards the lower powers, are more simple than the ordinary expressions in terms of x .

The expressions are continued by means of a recurring formula, viz. if

$$\begin{aligned} Sx^{2p-1} &= \frac{1}{2p-2} [a_1u^p - a_2u^{p-1} + \dots + (-)^{p-1}a_{p-1}u^2], \\ Sx^{2p+1} &= \frac{1}{2p} [u^p - b_1u^{p-1} + \dots + (-)^pb_pu^2], \end{aligned}$$

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then

$$\begin{aligned} 2p(2p-1)a_1 &= (2p-2)(2p-3)b_1 - p(p-1), \\ 2p(2p-1)a_2 &= (2p-2)(2p-3)b_2 - 4(p-1)(p-2)b_1, \\ 2p(2p-1)a_3 &= (2p-6)(2p-7)b_3 - (p-2)(p-3)b_2, \\ &\vdots \\ 2p(2p-1)a_{p-2} &= 5 \cdot 6 b_{p-2} - 3 \cdot 4 b_{p-3}, \\ 0 &= 3 \cdot 4 b_{p-2} - 2 \cdot 3 b_{p-3}, \end{aligned}$$

by means of which the coefficients b can be determined when the coefficients a are known.

Jacobi remarks also that the expressions for the sums of the even powers may be obtained from those for the odd powers by means of the formula

$$Sx^{2p} = \frac{1}{2p+1} \partial_u Sx^{2p+1},$$

which shows that any such sum will be of the form $(2x+1)u$ into a rational and integral function of u : thus in particular

$$Sx^2 = \frac{1}{6}(2x+1)u.$$

To show *a priori* that Sx^{2p+1} can be expressed as a rational and integral function of u , it may be remarked that $Sx^{2p+1} = \phi x$, where ϕx denotes the summatory integral $\Sigma(x+1)^{2p+1}$, taken so as to vanish for $x=0$: ϕx is a rational and integral function of x of the degree $2p+2$, and which, as is well known, contains x^2 as a factor. Suppose that y is any positive or negative integer less than x , we have

$$\phi x - \phi y = (y+1)^{2p+1} + (y+2)^{2p+1} + \dots + x^{2p+1},$$

and in particular putting $y = -1 - x$,

$$\phi x - \phi(-1-x) = (-x)^{2p+1} + (1-x)^{2p+1} + \dots + x^{2p+1} = 0,$$

since the terms destroy each other in pairs; we have therefore $\phi x = \phi(-1-x)$. Now $u = x^2 + x$, or writing this equation under the form $x^2 = -x + u$, we see that any rational and integral function of x may be reduced to the form $Px + Q$, where P and Q are rational and integral functions of u . Write therefore $\phi x = Px + Q$: the substitution of $-1-x$ in the place of x leaves u unaltered, and the equation $\phi(-1-x) = \phi x$ thus shows that $P=0$; we have therefore $\phi x = Q$, a rational and integral function of u . Moreover ϕx as containing the factor x^2 , must clearly contain the factor u^2 , and the expressions for Sx^{2p+1} are thus shown to be of the form given by Jacobi.

We may obtain a finite expression for Sx^n in terms of the differences of 0^n as follows: we have

$$Sx^n = 1^n + 2^n + \dots + x^n = \{(1+\Delta) + (1+\Delta)^2 + \dots + (1+\Delta)^x\} 0^n = \frac{1+\Delta}{\Delta} \{(1+\Delta)^x - 1\} 0^n,$$

and putting $(1 + \Delta)^x = e^{x \log(1 + \Delta)}$ and observing that the term independent of x vanishes, and that the terms containing powers higher than x^{n+1} also vanish, we have

$$Sx^n = S_k \left\{ \frac{1 + \Delta}{\Delta} \log^k(1 + \Delta) \right\} 0^n \cdot \frac{x^k}{\Pi k},$$

where the summation with respect to k , extends from $k = 1$ to $k = n + 1$, or what is the same thing (since the term corresponding to $k = 1$ in fact vanishes) from $k = 2$ to $k = n + 1$.

The equation $x^2 = -x + u$ gives

$$x^k = P_k x + Q_k,$$

and it is easy to see that writing for shortness

$$M_k = 1 + \frac{k-3}{1}u + \frac{k-4 \cdot k-5}{1 \cdot 2}u^2 + \frac{k-5 \cdot k-6 \cdot k-7}{1 \cdot 2 \cdot 3}u^3 + \dots,$$

$$N_k = \frac{k-1}{1} + \frac{k-3 \cdot k-4}{1 \cdot 2}u + \dots,$$

where the series is to be continued to the term $u^{\frac{1}{2}(k-2)}$ or $u^{\frac{1}{2}(k-3)}$ according as k is even or odd, we have

$$P_k = (-)^{k+1} M_{k-1}, \quad Q_k = (-)^k u M_k.$$

we have consequently

$$Sx^n = x S_k \left\{ \frac{1 + \Delta}{\Delta} \log^k(1 + \Delta) \right\} 0^n \cdot \frac{(-)^{k+1} M_{k-1}}{\Pi k} \\ + S_k \left\{ \frac{1 + \Delta}{\Delta} \log^k(1 + \Delta) \right\} 0^n \cdot \frac{(-)^k u M_k}{\Pi k}.$$

If n is odd, $= 2p + 1$, then (by what precedes) the first term vanishes, or we have

$$S_k \left\{ \frac{1 + \Delta}{\Delta} \log^k(1 + \Delta) \right\} 0^{2p+1} \cdot \frac{(-)^{k+1} M_{k-1}}{\Pi k} = 0, \quad (k = 1 \text{ to } k = 2p + 2),$$

and the formula becomes

$$Sx^{2p+1} = S_k \left\{ \frac{1 + \Delta}{\Delta} \log^k(1 + \Delta) \right\} 0^{2p+1} \cdot \frac{(-)^k u M_k}{\Pi k}, \quad (k = 1 \text{ to } k = 2p + 2),$$

which it may be noticed puts in evidence the factor u but not the factor u^2 .

If n is even, $= 2p$, then (by what precedes) the coefficient of x is to the constant term in the ratio $2 : 1$, or we have

$$S_k \left\{ \frac{1 + \Delta}{\Delta} \log^k(1 + \Delta) \right\} 0^{2p} \cdot \frac{(-)^k (M_{k-1} - 2u M_k)}{\Pi k} = 0, \quad (k = 1 \text{ to } k = 2p + 1),$$

and the formula becomes

$$Sx^{2p} = (2x + 1) S_k \left\{ \frac{1 + \Delta}{\Delta} \log^k(1 + \Delta) \right\} 0^{2p} \cdot \frac{(-)^k u M_k}{\Pi k}, \quad (k = 1 \text{ to } k = 2p + 1).$$

The values of the functions M are as follows:

$$\begin{aligned} M_1 &= 0, \\ M_2 &= 1, \\ M_3 &= 1, \\ M_4 &= 1 + u, \\ M_5 &= 1 + 2u, \\ M_6 &= 1 + 3u + u^2, \\ M_7 &= 1 + 4u + 3u^2, \\ &\&c. \end{aligned}$$

As a simple example of the formulae, we have

$$\begin{aligned} Sx^3 = & \left\{ \frac{1+\Delta}{\Delta} \log^2(1+\Delta) \right\} 0^3 \cdot \frac{1}{2}u \\ & - \left\{ \frac{1+\Delta}{\Delta} \log^3(1+\Delta) \right\} 0^3 \cdot \frac{1}{6}u \\ & + \left\{ \frac{1+\Delta}{\Delta} \log^4(1+\Delta) \right\} 0^3 \cdot \frac{1}{24}(u+u^2), \end{aligned}$$

and the coefficients are

$$\begin{aligned} (\Delta - \tfrac{1}{2}\Delta^2) 0^3 &= 1 - \tfrac{1}{2} \cdot 6 = 1, \\ (\Delta^2 - \tfrac{1}{2}\Delta^3) 0^3 &= 6 - 4 \cdot 6 = -3, \\ \Delta^4 0^3 &= 3 \cdot 3 = 6, \end{aligned}$$

and therefore

$$Sx^3 = \tfrac{1}{2}u - \tfrac{1}{2}u + \tfrac{1}{4}(u+u^2) = \tfrac{1}{2}u^2,$$

which is right; the example shows however that the calculation for the higher powers would be effected more readily by means of Jacobi's recurring formula.

2, Stone Buildings, 27th Oct., 1857.

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ON THE SYSTEM OF CONICS WHICH PASS THROUGH THE SAME FOUR POINTS.

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 206–207.]

I CONSIDER the system of conics passing through the same four points; these points may be real or imaginary, but it is assumed that there is a real system of conics, this will in fact be the case if two conics of the system are real. The four points are therefore given as the points of intersection of two real conics, and it will be proper to assume in the first instance that the conics intersect in four separate and distinct points, none of them at infinity. The four points may be all real, or two real and two imaginary, or all imaginary.

First, if the points are all real, we have here two cases, viz. each of the points may lie outside of the triangle formed by the other three, or as this may be expressed, the points may form a convex quadrangle; or else one of the points may be inside the triangle formed by the other three, or as this may be expressed, the points may form a triangle and interior point. In each case the pairs of lines joining the points, two and two together, will be conics (degenerate hyperbolas) forming part of the system of conics. Consider the two cases separately.

[Figure A: Four real points forming a convex quadrangle; pairs of lines through the four points and two parabolas dividing the plane into five regions, the central region containing ellipses and the four outer regions each containing hyperbolas.]

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Fig. A. Four real points forming a convex quadrangle. The system contains two parabolas, and the pairs of lines and the parabolas divide the plane of the figure into five distinct regions, one of which contains only ellipses, and the other four contain each of them hyperbolas.

Fig. A'. Four real points forming a triangle and interior point. The system does not contain any parabolas, the three pairs of lines divide the plane of the figure into three distinct regions, each of which contains only hyperbolas.

[Figure A': Triangle of three real points with an interior fourth point; the three pairs of lines divide the plane into three regions each containing hyperbolas.]

Next, if the points are two of them real and two of them imaginary. The line joining the two imaginary points will be real and this line may meet the line joining the two real points, in a point outside the two real points, or included between them, i.e. the real centre of the quadrangle may lie outside the real points, or may be included between them; I consider the two cases separately.

Fig. B. Two real and two imaginary points, the real centre of the quadrangle lying outside the real points. The system contains two parabolas, and these with the line joining the two real points and the line joining the two imaginary points divide the plane of the figure into three regions, one of which contains ellipses and the other two contain each of them hyperbolas.

[Figure B: Two real points joined by a line crossed by a vertical line (locus of the two imaginary points), with two parabolas, dividing the plane into three regions.]

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Fig. B'. Two real and two imaginary points, the real centre of the quadrangle lying between the real points. There are no parabolas, and the system contains only hyperbolas.

[Figure B': Two real points with the real centre of the quadrangle lying between them; no parabolas, only hyperbolas.]

Lastly, when the four points are imaginary. We have here only a single case.

Fig. C. Four imaginary points. The points lie on two real lines, there are (besides the point of intersection of these lines) two other real centres of the quadrangle, which lie harmonically with respect to the two lines. The system contains two parabolas and these and the two lines divide the plane of the figure into four regions, two of which contain each of them ellipses, and the other two contain each of them hyperbolas.

[Figure C: Four imaginary points represented by two crossing real lines and two parabolas in opposite quadrants, dividing the plane into four regions.]

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NOTE ON THE EXPANSION OF THE TRUE ANOMALY.

[From the *Quarterly Mathematical Journal*, vol. II. (1858), pp. 229–232.]

If the true anomaly and the mean anomaly are respectively denoted by u , m , and if e be the eccentricity, then as usual $u - e \sin u = m$; and if we write

$$\lambda = \frac{1 - \sqrt{1 - e^2}}{e},$$

and take c to denote the base of the hyperbolic system of logarithms, we have

$$u = m + 2 \sum_1^\infty A_r \frac{\sin rm}{r},$$

and

$$A_r = \lambda^r c^{\frac{1}{2}re(\lambda - \lambda^{-1})} + \lambda^{-r} c^{-\frac{1}{2}re(\lambda - \lambda^{-1})},$$

where, after expanding the exponentials, the negative powers of λ are to be rejected and the term independent of λ is to be multiplied by $\frac{1}{2}$ (see *Camb. Math. Journal*, t. I. [1839] p. 228 and t. III. [1843] p. 165, [4]).

It is easily seen that e^r is the lowest power of e which enters into the value of A_r , and the question arises to find the numerical coefficient of the term in question; this is readily obtained from the formula; in fact considering first a term of the form

$$\lambda^{-r} c^{\frac{1}{2}re \cdot s} (\lambda - \lambda^{-1})^s,$$

since λ is itself of the order e , when the negative powers of λ are rejected this is at least of the order e^s and it is consequently to be neglected if $s > r$. But if $s < r$ all the powers of λ are negative and the term is to be rejected. The only case to be

considered is therefore that of $s = r$, in which case there is a term containing e^r . We thus obtain from $\lambda^{-r} c^{\frac{1}{2}re \cdot r} (\lambda - \lambda^{-1})^r$ the term

$$\frac{1}{2^r} \cdot \frac{r^r e^r}{1 \cdot 2 \cdot 3 \dots r}.$$

In the next place a term of the form $\lambda^r c^{\frac{1}{2}re \cdot s} (\lambda - \lambda^{-1})^s$ is at least of the order e^s if $s > r$, or the terms to be considered are those for which $s = r$ or $s < r$. But in such term the only part of the order e^r is

$$(-)^s \lambda^{-s+r} c^{\frac{1}{2}re \cdot s},$$

or, since neglecting higher powers of e we have $\lambda = \frac{1}{2}e$, this is

$$(-)^s 2^{-r+s} r^s e^r,$$

and the set of terms arising from

$$\lambda^r c^{\frac{1}{2}re \cdot s} (\lambda - \lambda^{-1})^s$$

is

$$\frac{e^r}{2^r} \left\{ \frac{1}{1} + \frac{r}{1} + \frac{r^2}{1 \cdot 2} + \dots + \frac{r^{r-1}}{1 \cdot 2 \dots (r-1)} + \frac{1}{2} \cdot \frac{r^r}{1 \cdot 2 \dots r} \right\},$$

the last term being divided by 2 because arising from a term independent of λ . Hence the first term of A_r is

$$\frac{e^r}{2^r} \left\{ \frac{1}{1} + \frac{r}{1} + \frac{r^2}{1 \cdot 2} + \dots + \frac{r^r}{1 \cdot 2 \dots r} \right\},$$

a result which it may be remarked is contained in the general formula given in Hansen's Memoir "Entwicklung des Products u. s. w.," *Leipzig Trans.*, t. II. p. 277 (1853).

The preceding expression is

$$\frac{e^r c^r}{2^r} \cdot \frac{1}{\Gamma(r+1)} \int_0^\infty x^r c^{-x} dx,$$

and to find its value when r is large, we have

$$\begin{aligned} \int_0^\infty x^r c^{-x} dx &= \int_0^\infty (y+r)^r c^{-y-r} dy = r^r c^{-r} \int_0^\infty \left(1 + \frac{y}{r}\right)^r c^{-y} dy \\ &= r^r c^{-r} \int_0^\infty c^{-\frac{y^2}{2r} + \frac{y^3}{3r^2} - \&c.} dy \\ &= r^r c^{-r} \int_0^\infty c^{-\frac{y^2}{2r}} \left(1 + \frac{y^3}{3r^2} + \dots\right) dy \\ &= r^r c^{-r} \sqrt{2r} \left(1 + \frac{2\sqrt{2}}{3\sqrt{r}} + \dots\right) c^{-\frac{1}{2}} dz. \end{aligned}$$